

Parking Aid

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The parking aid sensors detect objects approximately 152 cm (60 in) from the rear of the vehicle, 50 cm (20 in) from the rear side of the vehicle, and 30 cm (12 in) above the ground. The Parking Aid Module (PAM) calculates the distance to an object within the 170 degree semicircular area around the rear of the vehicle.

A variable-rate warning tone is generated from a speaker attached to the PAM . The speaker increases its warning tone rate as the vehicle gets closer to an obstacle. When an object is detected within 30 cm (12 in) of the sensors, the warning tone becomes continuous. When the PAM sounds the parking aid tone, it sends a volume cutback enabled message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The BCM sends a message to the Audio Front Control Module (ACM) over the Medium Speed Controller Area Network (MS-CAN) . When the ACM receives the message, it reduces the speaker output so the parking aid tone can be heard.

For vehicles that have a base audio system that is not on the Medium Speed Controller Area Network (MS-CAN) , the audio speaker output does not reduce when the parking aid tone sounds.

The parking aid system is enabled when the ignition switch is in the RUN position and the transmission in reverse. The parking aid system is disabled if a fault is detected in 1 of the 4 sensors, the parking aid speaker, or the PAM . This is indicated by a parking aid disabled warning message in the message center.

The parking aid system is enabled and disabled through the message center, located in the Instrument Panel Cluster (IPC) .

If a MyKey® restricted key is in the ignition cylinder, the parking aid menu is not available in the message center and cannot be deactivated.

If the vehicle is equipped with an integrated electronic Trailer Brake Control (TBC) module and a trailer with electric trailer brakes is connected to the vehicle, the parking aid system is disabled.

The PAM communicates on the HS-CAN and can be diagnosed with a scan tool. The PAM communicates with the scan tool, the BCM , the PCM and the IPC .

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Body Control Module (BCM) fuse 34 (10A) • Wiring, terminals or connectors • Parking Aid Module (PAM) • Parking aid sensors • Parking aid speaker (integral to the PAM)

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
- 4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

- 5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

- 6. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.
- 7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record the continuous memory DTCs.
- 8. Clear the continuous DTCs and carry out the self-test diagnostics for the PAM .
- 9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
- 10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

PID Chart

PID	Description
LRI_DIST	Left rear inner sensor distance to object
LRO_DIST	Left rear outer sensor distance to object
RRI_DIST	Right rear inner sensor distance to object
RRO_DIST	Right rear outer sensor distance to object
TRANSGR	Transmission gearshift (PRNDL) position

DTC Charts

Parking Aid Module (PAM) DTC Chart

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens, or shorts to ground. Continuous Memory DTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action
B1B44:01	Right Rear Outer Sensor: General Electrical Failure	GO to Pinpoint Test B.
B1B44:96	Right Rear Outer Sensor: Component Internal Failure	GO to Pinpoint Test B.
B1B46:01	Right Rear Inner Sensor: General Electrical Failure	GO to Pinpoint Test B.
B1B46:96	Right Rear Inner Sensor: Component Internal Failure	GO to Pinpoint Test B.
B1B48:01	Left Rear Outer Sensor: General Electrical Failure	GO to Pinpoint Test B.
B1B48:96	Left Rear Outer Sensor: Component Internal Failure	GO to Pinpoint Test B.
B1B50:01	Left Rear Inner Sensor: General Electrical Failure	GO to Pinpoint Test B.
B1B50:96	Left Rear Inner Sensor: Component Internal Failure	GO to Pinpoint Test B.
B1B52:12	Rear Sounder-Park Aid: Circuit Short to Battery	CLEAR the DTCs. REPEAT the self-test. If DTC B1B52:12 is recorded again, INSTALL a new Parking Aid Module (PAM) . REFER to Parking Aid Module (PAM) in this section. REPEAT the self-test.
B1B52:14	Rear Sounder-Park Aid: Circuit Short to Ground or Open	CLEAR the DTCs. REPEAT the self-test. If DTC B1B52:14 is recorded again, INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. REPEAT the self-test.
U0155:87	Lost Communication With Instrument Panel Cluster (IPC) Control Module: Missing Message	GO to Pinpoint Test C.
U0294:87	Lost Communication With Powertrain Control Monitor Module: Missing Message	GO to Pinpoint Test D.

DTC	Description	Action
U3000:41	Control Module: General Check Sum Failure	CLEAR the DTCs. REPEAT the self-test. If DTC U3000:41 is recorded again. INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. CLEAR all DTCs. REPEAT the self-test.
U3000:46	Control Module: Calibration/Parameter Memory Failure	CLEAR the DTCs. REPEAT the self-test. If DTC U3000:46 is recorded again, INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. REPEAT the self-test.
U3000:54	Control Module: Missing Calibration	The PAM needs to be configured. CARRY OUT Programmable Module Installation (PMI) for the PAM . REFER to Section 418-01. CLEAR all DTCs. REPEAT the self-test.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
No communication with the Parking Aid Module (PAM)	- Fuse - Wiring, terminals or connectors - PAM	REFER to Section 418-00.
The parking aid is inoperative/does not operate correctly	- Parking aid sensor alignment - PAM - Rear bumper	GO to Pinpoint Test A.
Continuous or intermittent tone when no obstacles or fault codes are present (certain obstacles may not be detected or recognized due to their shape, size or material)	- Dirty or iced over parking aid sensor(s) - Parking aid sensor bezel(s) or parking aid sensor(s) locked into the rear bumper incorrectly - Incorrectly mounted sensor and/or grommet - Parking aid sensors are not aligned correctly	- CLEAN the rear bumper and sensors with high-pressure water. - REMOVE and correctly INSTALL the parking aid sensor(s) and or bezel(s) as necessary. REFER to Parking Aid Sensor in this section.

	• Parking aid sensor(s) • PAM	• CARRY OUT the azimuth system check and elevation system check. REFER to Azimuth System Check and Elevation System Check in this section.
• The audio does not reduce when the parking aid tone sounds	• Communication network concern • Instrument Panel Cluster (IPC) • Audio Front Control Module (ACM)	• RETRIEVE and REPAIR all related IPC and ACM DTCs. For IPC DTCs, REFER to Section 413-01. For ACM DTCs, Refer to the appropriate section in Group 415 for the procedure.
• The reverse parking aid menu is not available in the message center and cannot be deactivated	• A MyKey® restricted key is in use	• The parking aid disable feature is not available when a MyKey® restricted key is in use. This is normal operation.
• The message center displays the "Rear Parking Aid Off" selection message and cannot be selected ON	• The vehicle is equipped with a Trailer Brake Control (TBC) module and a trailer with electric trailer brakes is connected to the vehicle.	• This is normal operation. The parking aid is selected OFF when a Trailer Brake Control (TBC) module and a trailer with electric trailer brakes is connected.

Pinpoint Tests

Pinpoint Test A: The Parking Aid Is Inoperative/Does Not Operate Correctly

Normal Operation

The parking aid system is enabled when the ignition switch is in the RUN position and the reverse gear is selected. The Parking Aid Module (PAM) receives a reverse gear input message from the PCM over the High Speed Controller Area Network (HS-CAN) . The PAM supplies reference voltage and signal return to the parking aid sensors. The PAM receives an input signal from each individual parking aid sensor. The system can be disabled through the message center which is part of the Instrument Panel Cluster (IPC) .

This pinpoint test is intended to diagnose the following:

- Parking aid sensor alignment
- PAM
- Rear bumper

PINPOINT TEST A : THE PARKING AID IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
A1 CHECK THE DTCS FROM BOTH THE CONTINUOUS AND ON-DEMAND PAM SELF-TESTS	
• Check the PAM DTCs from the continuous and on-demand self-tests.	
Are any PAM DTCs recorded?	
Yes	REFER to DTC Charts in this section.

No	GO to A2 .
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A2 CHECK THE MESSAGE CENTER FOR CORRECT OPERATION

- While observing the message center, disable and enable the parking aid system.

Does the message center display REAR PARK AID OFF when the parking aid system is disabled, and display REAR PARK AID ON when the parking aid system is enabled?

Yes	GO to A3 .
No	REFER to Section 413-01.

A3 CHECK FOR CORRECT REVERSE GEAR INPUT

- Apply the parking brake.
- Ignition ON.
- Select REVERSE.
- Enter the following diagnostic mode on the scan tool: PAM DataLogger .
- Monitor the PAM TRANSGR PID.

Does the PID read REVERSE (transmission in reverse)?

Yes	GO to A4 .
No	GO to A5 .

A4 CHECK THE PARKING AID SENSORS FOR CORRECT ALIGNMENT

NOTE: *If the vehicle was in a rear end collision, the parking aid sensors may not be aligned correctly.*

- Check the bumper for correct alignment.
- Carry out the azimuth system check. Refer to Azimuth System Check in this section.
- Carry out the elevation system check. Refer to Elevation System Check in this section.

Does the parking aid system pass both system checks?

Yes	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.
No	REALIGN or INSTALL a new bumper cover. REFER to Section 501-19. If the vehicle was in a rear end collision, additional body work may be required to bring the parking aid sensors into correct alignment.

A5 CHECK FOR CORRECT PAM OPERATION

- Disconnect the PAM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PAM connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: DTCs B1B44:01, 96; B1B46:01, 96; B1B48:01, 96; B1B50:01, 96

Refer to Wiring Diagrams Cell [131](#) , Parking Aid for schematic and connector information.

Normal Operation

The Parking Aid Module (PAM) supplies a reference voltage and signal return to the parking aid sensors. When an object is present behind the vehicle, the reference voltage to the PAM changes based on how close the object is from the rear of the vehicle. The input signal from the sensors determines the distance of the object to the vehicle. A fault condition occurs when the PAM detects an open in the parking aid sensor circuits or the parking aid sensor input signal is shorted to ground.

DTC Description	Fault Trigger Conditions
• B1B44:01 — Right Rear Outer Sensor: General Electrical Failure	A continuous and on-demand DTC that sets when the right rear outer parking aid sensor circuits are shorted to battery voltage, open or the sensor signal line is shorted to ground.
• B1B44:96 — Right Rear Outer Sensor: Component Internal Failure	A continuous and on-demand DTC that sets when the PAM detects an internal failure to the right rear outer parking aid sensor.
• B1B46:01 — Right Rear Inner Sensor: General Electrical Failure	A continuous and on-demand DTC that sets when the right rear inner parking aid sensor circuits are shorted to battery voltage, open or the sensor signal line is shorted to ground.
• B1B46:96 — Right Rear Inner Sensor: Component Internal Failure	A continuous and on-demand DTC that sets when the PAM detects an internal failure to the right rear inner parking aid sensor.
• B1B48:01 — Left Rear Outer Sensor: General Electrical Failure	A continuous and on-demand DTC that sets when the left rear outer parking aid sensor circuits are shorted to battery voltage, open or the sensor signal line is shorted to ground.

DTC Description	Fault Trigger Conditions
B1B48:96 — Left Rear Outer Sensor: Component Internal Failure	A continuous and on-demand DTC that sets when the PAM detects an internal failure to the left rear outer parking aid sensor.
B1B50:01 — Left Rear Inner Sensor: General Electrical Failure	A continuous and on-demand DTC that sets when the left rear inner parking aid sensor circuits are shorted to battery voltage, open or the sensor signal line is shorted to ground.
B1B50:96 — Left Rear Inner Sensor: Component Internal Failure	A continuous and on-demand DTC that sets when the PAM detects an internal failure to the left rear inner parking aid sensor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PAM
- Parking aid sensor(s)
- Parking aid bumper harness

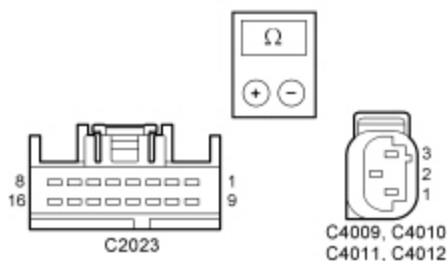
PINPOINT TEST B : DTCS B1B44:01, 96; B1B46:01, 96; B1B48:01, 96; B1B50:01, 96

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
B1 RETRIEVE THE DTCS FROM BOTH THE CONTINUOUS AND ON-DEMAND PAM SELF-TESTS	
Check the PAM DTCS from the continuous and on-demand self-tests.	
Are multiple DTCS recorded?	
Yes	GO to B2 .
No	For DTC B1B44:01, B1B44:96, B1B46:01, B1B46:96, B1B48:01, B1B48:96, B1B50:01 or B1B50:96, GO to B3 . For all other DTCS, REFER to DTC Charts in this section.
B2 CHECK THE BUMPER WIRING HARNESS	
Inspect the bumper wiring harness for opens, shorts, grounds, or corrosion.	
Is the bumper wiring harness OK?	
Yes	GO to B3 .
No	REPAIR or INSTALL a new bumper wiring harness. CLEAR the DTCS. REPEAT the self-test.
B3 CHECK THE SENSOR CIRCUITRY FOR OPENS	
- Ignition OFF. - Disconnect: Suspect Parking Aid Sensor C4009, C4010, C4011 or C4012.	

- Disconnect: PAM [C2023](#) .

Measure the resistance between the PAM [C2023](#), harness side and the suspect parking aid sensor, harness side as follows:

DTC	Rear Sensor	PAM Connector- Pin	Sensor Connector- Pin	Circuit
B1B44:01, 96	RH outer	C2023 Pin 9	C4011 Pin 1	VMP17 (YE/OG)
		C2023 Pin 13	C4011 Pin 3	RMP07 (GN/WH)
B1B46:01, 96	RH inner	C2023 Pin 10	C4012 Pin 1	VMP16 (YE/GY)
		C2023 Pin 13	C4012 Pin 3	RMP07 (GN/WH)
B1B48:01, 96	LH outer	C2023 Pin 12	C4009 Pin 1	VMP15 (YE/GN)
		C2023 Pin 13	C4009 Pin 3	RMP07 (GN/WH)
B1B50:01, 96	LH inner	C2023 Pin 11	C4010 Pin 1	VMP14 (WH/OG)
		C2023 Pin 13	C4010 Pin 3	RMP07 (GN/WH)



- N0102257

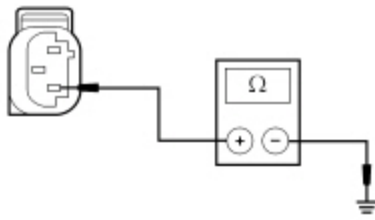
Are the resistances less than 5 ohms?

Yes	GO to B4 .
No	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

B4 CHECK THE SENSOR CIRCUITRY FOR SHORTS TO GROUND

- Measure the resistance between the suspect parking aid sensor, harness side and ground as follows:

DTC	Rear Sensor	Connector- Pin	Circuit
B1B44:01, 96	RH outer	C4011 Pin 1	VMP17 (YE/OG)
B1B46:01, 96	RH inner	C4012 Pin 1	VMP16 (YE/GY)
B1B48:01, 96	LH outer	C4009 Pin 1	VMP15 (YE/GN)
B1B50:01, 96	LH inner	C4010 Pin 1	VMP14 (WH/OG)



- N0102258

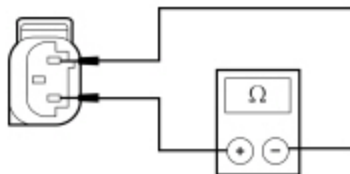
Is the resistance greater than 10,000 ohms?

Yes	GO to B5 .
No	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

B5 CHECK THE PARKING AID SENSOR CIRCUITRY FOR A SHORT TOGETHER

- Measure the resistance between the suspect parking aid sensor pins, harness side as follows:

DTC	Parking Aid Sensor	Connector- Pin/Circuit	Connector- Pin/Circuit
B1B44:01, 96	RH outer	C4011 Pin 1 VMP17 (YE/OG)	C4011 Pin 3 RMP07 (GN/WH)
B1B46:01, 96	RH inner	C4012 Pin 1 VMP16 (YE/GY)	C4012 Pin 3 RMP07 (GN/WH)
B1B48:01, 96	LH outer	C4009 Pin 1 VMP15 (YE/GN)	C4009 Pin 3 RMP07 (GN/WH)
B1B50:01, 96	LH inner	C4010 Pin 1 VMP14 (WH/OG)	C4010 Pin 3 RMP07 (GN/WH)



- N0120897

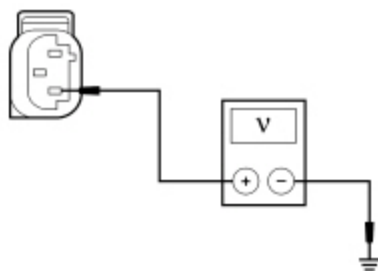
Is the resistance greater than 10,000 ohms?

Yes	GO to B6 .
No	REPAIR the circuits in question. CLEAR the DTCs. TEST the system for normal operation.

B6 CHECK THE SENSOR CIRCUITRY FOR SHORTS TO VOLTAGE

- Ignition ON.
- Measure the voltage between the suspect parking aid sensor, harness side and ground as follows:

DTC	Rear Sensor	Connector- Pin	Circuit
B1B44:01, 96	RH outer	C4011 Pin 1	VMP17 (YE/OG)
B1B46:01, 96	RH inner	C4012 Pin 1	VMP16 (YE/GY)
B1B48:01, 96	LH outer	C4009 Pin 1	VMP15 (YE/GN)
B1B50:01, 96	LH inner	C4010 Pin 1	VMP14 (WH/OG)



• N0082178

Is any voltage present?

Yes	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.
No	If all of the parking aid sensors recorded DTCs, GO to B7 . If one or more parking aid sensor(s) recorded DTCs, INSTALL a new sensor(s) for the one in question. REFER to Parking Aid Sensor in this section. CLEAR the DTCs. REPEAT the self-test.

B7 CHECK FOR CORRECT PAM OPERATION

- Disconnect the PAM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the PAM connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. TEST the system for normal operation.
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No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.
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Pinpoint Test C: DTC U0155:87

Normal Operation

- DTC U0155:87 (Lost Communication With Instrument Panel Cluster (IPC) Control Module: Missing Message) — the Parking Aid Module (PAM) set this DTC if data messages received from the Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds.

This pinpoint test is intended to diagnose the following:

- Module communication
- IPC
- PAM

PINPOINT TEST C : DTC U0155:87

C1 VERIFY CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to C2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

C2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the IPC pass the network test?

Yes	GO to C3 .
No	REFER to Section 418-00, The IPC Does Not Respond To The Scan Tool.

C3 CHECK THE IPC FOR DTCS

- Check for recorded IPC DTCs from the self-test.

Are any DTCs recorded?

Yes	REFER to Section 413-01.
No	GO to C4 .

C4 RECHECK THE PAM DTCS

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the ~~PAM~~ self-test.

Is DTC U0155:87 still present?

Yes	GO to C5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or intermittent fault condition.

C5 CHECK FOR DTC U0155 OR U0155:00 SET IN OTHER MODULES

- Ignition ON.
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0155 or U0155:00 set in the ~~ACM~~ or the ABS module?

Yes	INSTALL a new IPC . REFER to Section 413-01. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. TEST the system for normal operation.

Pinpoint Test D: DTC U0294:87

Normal Operation

- DTC U0294:87 (Lost Communication With Powertrain Control Monitor Module: Missing Message) — the Parking Aid Module (PAM) sets this DTC if data messages received from the PCM over the High Speed Controller Area Network (HS-CAN) are missing for 5 seconds.

This pinpoint test is intended to diagnose the following:

- Module communication
- ~~PAM~~
- PCM

PINPOINT TEST D : DTC U0294:87

D1 VERIFY CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to D2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

D2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the PCM pass the network test?

Yes	GO to D3 .
No	REFER to Section 418-00, The PCM Does Not Respond To The Scan Tool.

D3 CHECK THE PCM FOR DTCS

- Check for recorded PCM DTCS from the self-test.

Are any DTCS recorded?

Yes	REFER to Section 303-14.
No	GO to D4 .

D4 RECHECK THE PAM DTCS

- Clear the DTCS.
- Wait 10 seconds.
- Repeat the ~~PAM~~ self-test.

Is DTC U0294:87 still present?

Yes	GO to D5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or intermittent fault condition.

D5 CHECK FOR DTC U0100 OR U0100:00 SET IN OTHER MODULES

- Ignition ON.
- Retrieve the continuous memory DTCS from all modules.

Is DTC U0100 or U0100:00 set in the ABS module or the IPC ?

Yes	INSTALL a new PCM. REFER to Section 303-14. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new PAM . REFER to Parking Aid Module (PAM) in this section. TEST the system for normal operation.